# Glossary of terms

Part 1 lists the words of each session. Part 2 lists the industry terms by area. Learn the industry term. The plain words explain it.

## Part 1 The words of the sessions

| Term | Plain meaning | Session |
|---|---|---|
| **Activation function (sigmoid, ReLU)** | The bend applied to the weighted sum. Sigmoid gives a score from 0 to 1. ReLU gives 0 below zero and the total above it. | 20 |
| **Aggregate function, GROUP BY and HAVING** | COUNT, SUM, AVG, MIN and MAX reduce records to one value. HAVING filters the groups that GROUP BY makes. | 10 |
| **Artificial neuron (perceptron)** | A small calculation: a weighted sum of the inputs plus a bias, passed through an activation function. | 20 |
| **Bag of words and TF-IDF** | Bag of words counts each word in each message and ignores the order. TF-IDF weights those counts so rare words count more. | 21 |
| **Baseline model** | The simplest possible model, such as always predicting the majority class (DummyClassifier). Every real model has to beat it. | 14 |
| **Baseline model (seasonal naive forecast)** | The simplest forecast: copy the same month from last year. A model must beat it to be worth using. | 19 |
| **Binary classification (positive class, negative class)** | Predicting one of two categories. The positive class is the outcome we look for: default = 1. The negative class is the other: repaid = 0. | 15 |
| **Boolean expression** | A test that gives True or False, built with comparison operators such as >= and logical operators such as and. | 6 |
| **Boolean mask (Boolean indexing)** | A column of True and False values, one per row, used to keep only the rows marked True. | 7 |
| **Cardinality (one-to-many, many-to-one)** | How many records on each side match one key value. It decides if a merge keeps, loses or multiplies rows. | 9 |
| **Class imbalance (accuracy paradox)** | One class is far more common than the other, so a model that ignores the rare class can still have high accuracy. | 14 |
| **Coefficient and intercept** | The coefficient is the change in the prediction for one more unit of a predictor, with the other predictors held fixed. The intercept is the prediction when every predictor is 0. | 13 |
| **Conditional statement (if, elif, else)** | Code that runs one branch, chosen by a test that is True or False. It is the IF function written as code. | 6 |
| **Confidence interval (CI)** | A range built from the data. A 95 percent interval is the estimate plus or minus 1.96 standard errors. | 12 |
| **Confounder** | A variable that influences both of two variables you are comparing, so they look linked. Also called a confounding variable. | 4 |
| **Confusion matrix (TP, FP, FN, TN)** | A table that counts every kind of right and wrong prediction: true positives, false positives, false negatives and true negatives. | 16 |
| **Control total** | A total that comes from a trusted source, such as the till system, used to test your own total. | 8 |
| **Correlation (Pearson r)** | A number from -1 to +1 for how closely two numeric variables rise and fall together. Correlation is not causation. | 4 |
| **Cosine similarity and semantic search** | A score from -1 to 1 for how alike two vectors are. Semantic search ranks messages by that score. | 21 |

| Term | Plain meaning | Session |
|---|---|---|
| **Count and rate** | A count is how many. A rate divides the count by what could have happened, such as returns per 100 lines sold. | 11 |
| **Course project (capstone)** | The larger piece of work that ends a course. Yours is a question, data, a model beside a baseline model, its limits and a talk. You keep it. | 24 |
| **Data cleaning log** | A table of every change to the data, with the row counts before and after and the reason. It is the audit trail of the cleaning. | 8 |
| **Data drift (and concept drift)** | Data drift: the inputs change after training. Concept drift: the link between the inputs and the outcome changes. | 23 |
| **Data leakage (target leakage)** | A feature that contains the answer, or information only known after the outcome. It makes a model look brilliant and fail in real use. | 16 |
| **Data science** | Using data, tools and careful thinking to answer a question and support a decision. It is a way of working that runs from question to decision. | 1 |
| **Data type (int, float, str, bool)** | How Python stores a value: a whole number, a decimal number, text (a string) or True or False. | 5 |
| **DataFrame** | A table held in Python memory, with labelled rows and named columns. In Excel words: one tab of a workbook. | 7 |
| **Dataset** | A collection of data about one subject, arranged in tables. In Excel words: a workbook in which each tab is one table. | 1 |
| **Decision threshold (cut-off)** | The probability at or above which we predict the positive class. The default of 0.5 is rarely right for a business. | 15 |
| **Decision tree (root node, leaf node)** | A model made of yes or no questions. Each path ends in a leaf node, and the leaf gives the predicted probability. | 15 |
| **Dictionary (key-value pairs)** | Values stored under labels called keys. You look a value up by its key, as VLOOKUP does. | 5 |
| **Edge case (boundary value) and test case** | An edge case sits exactly where a rule changes. A test case pairs an input with the output you expect. | 6 |
| **Effect size** | How big the gap is, in units a person can feel, such as shillings or percent. Practical significance asks whether that size matters. | 12 |
| **Embedding (vector)** | A list of numbers that stands for the meaning of a message. Messages with similar meanings get similar vectors. | 21 |
| **Ensemble (random forest, gradient boosting)** | Many small models combined into one. A forest averages trees (bagging). Boosting adds trees, each fitting the residual errors so far. | 17 |
| **Evaluation set (precision at k, recall at k)** | A labelled list of test questions with the right passage. Precision at k and recall at k score the search. | 22 |
| **Exception (traceback)** | The error Python raises when a line cannot run. The traceback is its report. Its last line names the exception. | 5 |
| **Executive summary** | A short opening paragraph with the answer, the evidence, the main limit and the decision you ask for. A busy reader may read only this. | 24 |
| **Exploratory data analysis (EDA)** | A repeatable routine of looks, summaries and charts that you run on a new dataset before you model it or report on it. | 11 |
| **Fairness audit (proxy variable, disparate impact)** | Compare approval and error rates across groups. A proxy variable stands in for a column you left out. Disparate impact is the ratio of two approval rates. | 23 |
| **False positive (Type I error)** | Declaring a difference when there is none. Running many tests, or stopping early, makes it far more likely. | 12 |
| **Fan-out (row multiplication) and lost rows** | Fan-out: a merge repeats a record, so totals inflate. Lost rows: records with no match that an inner join drops. | 9 |

| Term | Plain meaning | Session |
|---|---|---|
| **Feature scaling and log transform** | Standardisation (z-score) puts variables on one scale. A log transform squeezes a long right tail. | 18 |
| **Features and target** | Features are the input variables. The target is the variable we predict. Statistics says independent and dependent variable. | 14 |
| **Feedforward neural network (multilayer perceptron, MLP)** | Neurons arranged in an input layer, hidden layers and an output layer, with values moving forward only. | 20 |
| **Figure and Axes (matplotlib)** | The Figure is the whole page. The Axes is the plot area on that page where the data are drawn and labelled. | 11 |
| **Finding** | One sentence that states a number, a comparison and a limit. | 11 |
| **Function (parameter, argument, return value)** | A named set of steps. The parameter is the name for the value that comes in, the argument is the value you pass, and the return value is the answer. | 6 |
| **Gradient descent and backpropagation** | Training: repeated small steps that lower the loss function. Backpropagation works out how much each weight changed the loss (its gradient). | 20 |
| **Granularity (grain)** | The level of detail of a table: one record per order, per order line or per branch per week. | 2 |
| **groupby and aggregation** | Split records into groups by a key, then reduce each group to one number, such as a sum or a mean. | 9 |
| **Hallucination** | Fluent text that no source supports. It can read well and still be false. | 22 |
| **Hyperparameter** | A setting chosen before training, such as max_depth. Parameters are learned from the data. Tuning tries combinations with cross-validation. | 17 |
| **Imputation** | Filling a missing value with a stated value, such as the median or a constant. Keep a flag to show which values were filled. | 8 |
| **Index** | The row labels down the left edge of a DataFrame. A fresh file is labelled 0, 1, 2 and so on. | 7 |
| **Indicator variable (dummy variable)** | A variable that is 1 when a record belongs to a category and 0 otherwise. One-hot encoding makes one for each level. | 13 |
| **Iteration (for loop)** | Repeating the same steps for each item in a list. It is fill down written as code. | 6 |
| **JOIN (INNER, LEFT)** | Combines two tables on a key. INNER keeps matches only. LEFT keeps every left record. | 10 |
| **k-means clustering** | Splits records into k clusters around k centroids. Inertia measures how tight they are. The elbow method suggests k. | 18 |
| **Lag feature** | An earlier value placed in the same row, such as last month's cases. It is built inside each district. | 19 |
| **Lifecycle (CRISP-DM)** | The phases a data project passes through. CRISP-DM is the industry reference: six phases, from business understanding to deployment. | 1 |
| **Limitation** | Something your result cannot tell you, such as who the data left out. You say it before you show the result. | 24 |
| **List** | Values in order inside square brackets. Each has an index, and the first index is 0. It is like one column of a sheet. | 5 |
| **Logistic regression (sigmoid, log-odds, regularisation)** | A model that bends a score from the features into a probability between 0 and 1 with the sigmoid. scikit-learn penalises large coefficients (regularisation, C = 1.0). | 15 |
| **Machine learning (ML)** | A computer finds a rule from examples. A person does not type the rule. | 1 |
| **Mean, median and mode** | Three ways to say what is typical. The mean is the sum divided by the count. The median is the middle value. The mode is the most common value. | 3 |

| Term | Plain meaning | Session |
|---|---|---|
| **Merge (join) and join key** | Combine two tables by matching a shared variable. The join type says which unmatched records survive. | 9 |
| **Missing value (NaN)** | A cell with no value. pandas shows it as NaN, SQL as NULL, and Excel as a blank cell. | 7 |
| **Missing value (NaN, NULL)** | A cell with no value. Ask whether it means unknown, not applicable or something real. Informative missingness: the blank itself is a fact, such as a walk-in shopper. | 2 |
| **Model** | A rule found from data that turns inputs into a prediction or an explanation. | 1 |
| **Model serialisation (persistence)** | Saving a trained model to a file, for example with joblib or pickle, so it can be loaded later without training again. | 23 |
| **Monte Carlo simulation** | Many random draws done by computer to see what a statistic does. In Sheets: RAND and INDEX, filled down. | 4 |
| **Null hypothesis (H0) and p-value** | H0 says there is no real difference. The p-value says how often chance alone would give a gap at least this big if H0 were true. | 12 |
| **NULL (three-valued logic)** | A missing value. A test on NULL is neither true nor false but unknown, so use IS NULL. | 10 |
| **One-hot encoding** | Turning a text variable into one 0 or 1 indicator column per category, so a model can use it. | 15 |
| **Outlier (IQR rule)** | A value far from the rest. The IQR rule flags values more than 1.5 times the interquartile range beyond the quartiles. | 8 |
| **Overfitting and underfitting** | Overfitting: the model fits noise, so training rows score well and new rows score worse. Underfitting: too simple to see the pattern. | 17 |
| **Parsimony (model selection)** | When scores are close, ship the simpler model. It is easier to explain, check and maintain. | 17 |
| **Permutation importance** | The drop in test score when one variable is shuffled. It shows what the model relies on. It does not show a cause. | 17 |
| **Personal data (data subject, data controller)** | Information that identifies a person. The person is the data subject. The organisation that decides how it is used is the data controller. | 23 |
| **Pipeline (scikit-learn)** | Preprocessing steps and a model packed into one object, so training and test records go through the same steps. | 19 |
| **Pivot table (GROUP BY)** | A table that groups records and summarises each group with an aggregation such as SUM, COUNT or AVERAGE (MEDIAN in Sheets and pandas). In SQL and pandas it is GROUP BY. | 3 |
| **pivot_table (wide and long format)** | A summary grid with one variable down the side and one across the top, which is wide format. | 9 |
| **Population and sample** | The population is every record you care about. A sample is the part you actually look at. The gap between a sample statistic and the population value is sampling error. How much a statistic changes from sample to sample is sampling variability. | 4 |
| **Portfolio** | A small set of your projects, for example GitHub repositories with a README, that a stranger can open and judge. | 24 |
| **Precision, recall and F1** | Precision: of the loans flagged, the share that defaulted. Recall: of the loans that defaulted, the share flagged. F1 combines both. | 16 |
| **Primary key and foreign key** | A primary key never repeats and names one record. A foreign key holds the primary key of another table and links the two. | 2 |
| **Principal component analysis (PCA)** | A dimensionality reduction method: it mixes many variables into a few principal components that keep most of the variance. | 18 |

| Term | Plain meaning | Session |
|---|---|---|
| **Project checklist (rubric)** | A list of the qualities that good work has, one row for each. You use it on your own project and to give a partner feedback. Nothing is scored. | 24 |
| **Prompt (system prompt, zero-shot, few-shot)** | The instruction you give a model. A system prompt sets rules. Zero-shot gives no examples. Few-shot gives a few worked examples. | 22 |
| **Quartiles and IQR** | Quartiles cut sorted values into four equal groups. The interquartile range (IQR) is Q3 minus Q1: the spread of the middle half. | 3 |
| **Query and clause** | A query is one question in SQL. A clause is one part of it, such as SELECT, WHERE or ORDER BY. | 10 |
| **R squared (coefficient of determination)** | The share of the variation in the target that the model accounts for, from 0 to 1. | 13 |
| **Reconciliation difference** | The gap between your cleaned total and the control total. Explain it, or find the defect behind it. | 8 |
| **Record (observation, instance, row)** | One row of a table: one thing the table describes. In the orders tab each record is one till order. | 2 |
| **Regression (linear regression model)** | A model that predicts a numeric target as an intercept plus a weighted sum of predictors. Ordinary least squares (OLS) finds the weights. | 13 |
| **Relational database (SQLite)** | Several tables linked by keys, stored in one file. SQLite is a free database that runs inside Python. | 10 |
| **Reproducibility** | A colleague who runs your notebook gets your result. It needs a fixed random seed, recorded library versions and a run from the top. | 23 |
| **Residual (prediction error)** | Actual value minus predicted value for one record. OLS minimises the sum of the squared residuals. | 13 |
| **Retrieval-augmented generation (RAG)** | Find the relevant chunks of your documents first (retrieval and ranking), then ask the model to answer from them (grounding). | 22 |
| **RFM analysis** | Three numbers per customer: recency (days since the last order), frequency (number of orders) and monetary value (total spend). | 18 |
| **ROC curve and AUC** | The ROC curve plots the true positive rate (recall, sensitivity) against the false positive rate at every threshold. AUC, the area under it, is 0.5 for a coin flip and 1.0 for a perfect model. | 16 |
| **Seasonality (with trend and noise)** | A pattern that comes back at the same time every year, such as a December peak. | 11 |
| **Series** | One column of a DataFrame. It has a name, a data type and the same index as the table. | 7 |
| **Skewness** | How lopsided a distribution is. Right-skewed means a long tail of large values on the right. | 3 |
| **Standard deviation** | A typical distance from the mean: the square root of the average squared distance (n minus 1 for a sample). | 3 |
| **Standard error** | The standard deviation of a statistic across many samples, such as the sample mean. It says how far one sample can sit from the truth. | 4 |
| **Standard error (SE)** | How much a sample mean varies from one random sample to the next. It is the standard deviation divided by the square root of the sample size. | 12 |
| **Stratified k-fold cross-validation** | Test the model k times, each time on a different fold, with every fold keeping the same share of defaults. Average the scores. Use it to tune before you test. | 16 |
| **Supervised learning** | A model learns the link between features (inputs) and a known target (the answer) from past records, then predicts the target for new records. | 14 |

| Term | Plain meaning | Session |
|---|---|---|
| **Text classification (sentiment analysis)** | A supervised model that gives each message a label, such as positive, neutral or negative. | 21 |
| **Text preprocessing (tokenisation, stop words)** | Cleaning raw text into tokens: lowercase it, remove noise with a regular expression, split it into words and drop filler words. | 21 |
| **Time series (trend, seasonality, noise)** | One variable recorded at regular times, such as cases each month. It splits into a slow trend, a yearly seasonality and leftover noise. | 19 |
| **Time-based split (temporal split)** | Train on the months before a cut date and test on the months after it. Never split a time series at random. | 19 |
| **Token and context window** | A token is a word or piece of a word that a language model reads and writes. The context window is how many tokens it can read at once. | 22 |
| **Training set and test set** | The training set is what the model learns from. The test set is held back and used once at the end to score the model. | 14 |
| **Transfer learning (pretrained model)** | Start from a network someone else trained and train only a new last layer on your own data. | 20 |
| **Unsupervised learning (for example clustering, anomaly detection)** | Learning with no answer column. The algorithm looks for groups or odd rows by itself. | 18 |
| **Variable (assignment)** | A name that stands for a stored value, like a named cell in Excel. In Python you choose the name, such as price_p001. In Session 2 a variable was a column of a table. Here it is a name that holds a value. | 5 |
| **Variable type and data type** | Variable type: how a variable is measured (numeric, categorical, datetime, text). Data type: how the cell stores it (integer, float, string). They are different ideas. | 2 |

## Part 2 Industry terms by area

The proper name comes first. Other names for the same thing are in brackets. Where a term is also one of the words of a session, the session number is shown.

### Data, tables and statistics

| Term | Plain meaning | Session |
|---|---|---|
| **A/B test (controlled experiment: treatment group and control group)** | A controlled experiment that compares a treatment group with a control group. | |
| **Aggregation** | Split records into groups by a key, then reduce each group to one number, such as a sum or a mean. | 9 |
| **Alternative hypothesis (H1)** | H1: there is a real difference. It is what you decide for if the data reject H0. | |
| **Bootstrap (resampling with replacement)** | Redraw samples with replacement from your own data many times to see how a statistic varies. | |
| **Causation** | One thing makes another happen. A correlation alone does not show it. | |
| **Composite key** | A key made of two or more variables that are unique only together, such as branch and week. | |
| **Confidence interval (CI)** | A range built from the data. A 95 percent interval is the estimate plus or minus 1.96 standard errors. | 12 |
| **Confidence level** | The 95 percent in a 95 percent confidence interval: how often such intervals catch the true value over many samples. | |
| **Confounder (confounding variable)** | A variable that influences both of two variables you are comparing, so they look linked. Also called a confounding variable. | 4 |
| **Control** | The group that does not get the change. It is the comparison. | |
| **Correlation (Pearson correlation coefficient r)** | A number from -1 to +1 for how closely two numeric variables rise and fall together. Correlation is not causation. | 4 |
| **Count** | A count is how many. A rate divides the count by what could have happened, such as returns per 100 lines sold. | 11 |
| **Crow's foot** | the three-prong line end on an ERD that marks the many side | |
| **Data cleaning log (an audit trail of each change with row counts)** | A table of every change to the data, with the row counts before and after and the reason. It is the audit trail of the cleaning. | 8 |
| **Data type (how the value is stored: integer, float, string, datetime)** | Variable type: how a variable is measured (numeric, categorical, datetime, text). Data type: how the cell stores it (integer, float, string). They are different ideas. | 2 |
| **Data type versus variable type** | data type: how the cell stores the value (string, integer, float). Variable type: how the value is measured (numeric, categorical, datetime, text). | |
| **Decomposition** | Splitting a time series into trend, seasonality and noise. | |
| **Effect size** | How big the gap is, in units a person can feel, such as shillings or percent. Practical significance asks whether that size matters. | 12 |
| **Entity relationship diagram (ERD)** | A drawing of the tables and the keys that link them. Crow's foot ends mark the many side. | |
| **Exploratory data analysis (EDA)** | A repeatable routine of looks, summaries and charts that you run on a new dataset before you model it or report on it. | 11 |
| **False negative (Type II error)** | Missing a real difference or a real case (Type II error). | |
| **False positive (Type I error)** | Declaring a difference when there is none. Running many tests, or stopping early, makes it far more likely. | 12 |

| Term | Plain meaning | Session |
|---|---|---|
| **Fan-out (row multiplication from a one-to-many join)** | Fan-out: a merge repeats a record, so totals inflate. Lost rows: records with no match that an inner join drops. | 9 |
| **Foreign key** | A primary key never repeats and names one record. A foreign key holds the primary key of another table and links the two. | 2 |
| **Granularity (data warehouse alias: grain)** | The level of detail of the table: one row per order, per order line or per branch per week. | 2 |
| **GROUP BY** | A table that groups records and summarises each group with an aggregation such as SUM, COUNT or AVERAGE (MEDIAN in Sheets and pandas). In SQL and pandas it is GROUP BY. | 3 |
| **groupby** | Split records into groups by a key, then reduce each group to one number, such as a sum or a mean. | 9 |
| **Identifier** | a variable that names a record (order_id). Never add it up. | |
| **Imputation (mean, median, constant imputation)** | Filling a missing value with a stated value, such as the median or a constant. Keep a flag to show which values were filled. | 8 |
| **Informative missingness (add a missing-indicator flag)** | A blank that carries meaning, such as a walk-in shopper with no customer id. Add a missing-indicator flag. | |
| **Inner join, left join, right join, full outer join** | Merge types. Inner keeps matches only, left keeps every left record, right keeps every right record, full outer keeps all. | |
| **Instalment-to-income ratio (payment-to-income ratio)** | The loan instalment divided by the monthly income. It shows how heavy the repayment is for the borrower. | |
| **Interquartile range (IQR)** | Q3 minus Q1: the spread of the middle half of the sorted values. | |
| **Join log** | An audit trail of the row counts before and after a join, so that a fan-out cannot go unseen. | |
| **Junction table (bridge table)** | a table that links two many-to-many tables | |
| **Lift (treatment mean minus control mean)** | The treatment mean minus the control mean. | |
| **Line items (the order lines table)** | The lines of an order, one record per product on the order (the order_lines table). | |
| **Many-to-many relationship (cardinality)** | Each record on one side can match many on the other side and the reverse. A junction table links them. | |
| **Many-to-one** | How many records on each side match one key value. It decides if a merge keeps, loses or multiplies rows. | 9 |
| **Many-to-one merge (pandas `validate='m:1'`)** | A merge where many left records match one lookup record. pandas checks it with validate="m:1". | |
| **Mean (arithmetic mean)** | Three ways to say what is typical. The mean is the sum divided by the count. The median is the middle value. The mode is the most common value. | 3 |
| **Median** | Three ways to say what is typical. The mean is the sum divided by the count. The median is the middle value. The mode is the most common value. | 3 |
| **Missing value (NaN in pandas, NULL in SQL)** | Say whether it means "unknown", "not applicable" or "a walk-in customer". | 2 |
| **Mode** | Three ways to say what is typical. The mean is the sum divided by the count. The median is the middle value. The mode is the most common value. | 3 |

| Term | Plain meaning | Session |
|---|---|---|
| **Monte Carlo simulation (random sampling)** | Many random draws done by computer to see what a statistic does. In Sheets: RAND and INDEX, filled down. | 4 |
| **Multiple comparisons problem** | The more tests you run, the more likely one gives a false positive by chance. | |
| **Noise (the decomposition of a time series)** | One variable recorded at regular times, such as cases each month. It splits into a slow trend, a yearly seasonality and leftover noise. | 19 |
| **Null hypothesis (H0)** | H0 says there is no real difference. The p-value says how often chance alone would give a gap at least this big if H0 were true. | 12 |
| **One-to-many** | How many records on each side match one key value. It decides if a merge keeps, loses or multiplies rows. | 9 |
| **Optional stopping (a form of p-hacking)** | Peeking at results and stopping when they look good. It raises false positives. | |
| **Orphan record** | a foreign key value with no matching primary key in the other table | |
| **Outlier (IQR rule, z-score rule)** | A value far from the rest. The IQR rule flags values more than 1.5 times the interquartile range beyond the quartiles. | 8 |
| **Percentage** | A part divided by the whole, times 100. | |
| **P-hacking** | Trying tests or stopping points until a p-value falls under the line. | |
| **Pivot table** | A table that groups records and summarises each group with an aggregation such as SUM, COUNT or AVERAGE (MEDIAN in Sheets and pandas). In SQL and pandas it is GROUP BY. | 3 |
| **Population** | The population is every record you care about. A sample is the part you actually look at. The gap between a sample statistic and the population value is sampling error. How much a statistic changes from sample to sample is sampling variability. | 4 |
| **Practical significance** | The effect is big enough to matter for the decision. Judge it with the effect size. | |
| **Primary key** | A primary key never repeats and names one record. A foreign key holds the primary key of another table and links the two. | 2 |
| **P-value** | How often chance alone would give a gap at least this big if H0 were true. | |
| **Random sample** | Records drawn by chance so that each has a known chance of being chosen. | |
| **Rate (events per 100 items)** | A count is how many. A rate divides the count by what could have happened, such as returns per 100 lines sold. | 11 |
| **Ratio** | One quantity divided by another, such as returns per order. | |
| **Reconciliation difference (never call it a residual: residual is a model word)** | The gap between your cleaned total and the control total. Explain it, or find the defect behind it. | 8 |
| **Record (observation, instance, row)** | One row of a table: one thing the table describes, such as one till order. | 2 |
| **Relational model** | Data kept in several tables that link through keys, as the tabs of the Lakeside workbook do. | |
| **Sample** | The population is every record you care about. A sample is the part you actually look at. The gap between a sample statistic and the population value is sampling error. How much a statistic changes from sample to sample is sampling variability. | 4 |
| **Sampling distribution** | The spread of a statistic, such as the mean, over many repeated samples. Its standard deviation is the standard error. | |

| Term | Plain meaning | Session |
|---|---|---|
| **Sampling error** | The gap between a sample figure and the true value for the whole population. | |
| **Sampling variability** | How much a sample figure changes from one random sample to the next. | |
| **Sampling with replacement** | Each record drawn goes back before the next draw, so it can be drawn again. | |
| **Seasonality** | A pattern that comes back at the same time every year, such as a December peak. | 11 |
| **Selection bias** | The records you see were chosen by something other than chance, so they misrepresent the whole. | |
| **Significance level (alpha)** | Alpha: the p-value below which you reject H0, chosen before the test, often 0.05. | |
| **Skewness (right-skewed, long right tail)** | How lopsided a distribution is. Right-skewed means a long tail of large values on the right. | 3 |
| **Standard deviation (sample standard deviation, ddof=1 in pandas)** | A typical distance from the mean: the square root of the average squared distance (n minus 1 for a sample). | 3 |
| **Standard error** | The standard deviation of a statistic across many samples, such as the sample mean. It says how far one sample can sit from the truth. | 4 |
| **Statistical power** | The chance that a test detects a real effect of a stated size. | |
| **Statistical significance** | The p-value is below the significance level, so chance alone is an unlikely explanation. | |
| **Survivorship bias** | You only see the records that survived a filter, such as the shops still open. | |
| **Treatment** | The group that gets the change being tested. | |
| **Trend** | One variable recorded at regular times, such as cases each month. It splits into a slow trend, a yearly seasonality and leftover noise. | 19 |
| **Unit of observation (what one record stands for)** | What one record stands for, such as one till order or one loan. | |
| **Uplift** | Another name for lift: the gain caused by the treatment. | |
| **Variable (feature or field; an attribute of each record)** | In a spreadsheet it is a column. In statistics it is a variable. In machine learning it is a feature (input) or the target. | 5 |
| **Variable type (how it is measured: numeric, categorical, ordinal, datetime, text)** | Variable type: how a variable is measured (numeric, categorical, datetime, text). Data type: how the cell stores it (integer, float, string). They are different ideas. | 2 |
| **Variance** | The average squared distance from the mean. Its square root is the standard deviation. | |
| **Without replacement** | Each record drawn stays out, so no record is drawn twice. | |

## Machine learning workflow, models and evaluation

| Term | Plain meaning | Session |
|---|---|---|
| **Accuracy** | The share of all predictions that are right. It misleads when one class is rare. | |
| **Accuracy paradox** | One class is far more common than the other, so a model that ignores the rare class can still have high accuracy. | 14 |
| **Activation function (sigmoid, ReLU)** | The bend applied to the weighted sum. Sigmoid gives a score from 0 to 1. ReLU gives 0 below zero and the total above it. | 20 |

| Term | Plain meaning | Session |
|---|---|---|
| **Adjusted R squared** | R squared corrected for the number of predictors, so extra useless predictors do not raise it. | |
| **Anomaly detection (outlier detection; IsolationForest)** | Learning with no answer column. The algorithm looks for groups or odd rows by itself. | 18 |
| **AUC** | The ROC curve plots the true positive rate (recall, sensitivity) against the false positive rate at every threshold. AUC, the area under it, is 0.5 for a coin flip and 1.0 for a perfect model. | 16 |
| **Backpropagation** | Training: repeated small steps that lower the loss function. Backpropagation works out how much each weight changed the loss (its gradient). | 20 |
| **Bagging** | Bootstrap aggregation: train many models on bootstrap samples and combine their votes. | |
| **Baseline model (majority-class baseline, DummyClassifier)** | The simplest possible model, such as always predicting the majority class (DummyClassifier). Every real model has to beat it. | 14 |
| **Batch** | A group of training records used for one correction step. | |
| **Bias** | The constant a neuron adds to its weighted sum. It is the neuron's intercept. | |
| **Bias-variance trade-off** | A model that is too simple misses the pattern (bias). One that is too flexible follows noise (variance). | |
| **Binary classification** | Predicting one of two categories. The positive class is the outcome we look for: default = 1. The negative class is the other: repaid = 0. | 15 |
| **Class imbalance (the accuracy paradox)** | One class is far more common than the other, so a model that ignores the rare class can still have high accuracy. | 14 |
| **Clusters** | Groups of similar records found by k-means or another clustering method. | |
| **Coefficient (slope for a numeric predictor)** | The coefficient is the change in the prediction for one more unit of a predictor, with the other predictors held fixed. The intercept is the prediction when every predictor is 0. | 13 |
| **Confusion matrix** | A table that counts every kind of right and wrong prediction: true positives, false positives, false negatives and true negatives. | 16 |
| **Customer segments** | Clusters of customers described in business words. | |
| **Data leakage (target leakage, train-test contamination)** | A feature that contains the answer, or information only known after the outcome. It makes a model look brilliant and fail in real use. | 16 |
| **Decision threshold (classification threshold)** | The probability at or above which we predict the positive class. The default of 0.5 is rarely right for a business. | 15 |
| **Decision tree (root node, internal node, leaf node, split)** | A model made of yes or no questions. Each path ends in a leaf node, and the leaf gives the predicted probability. | 15 |
| **Dependent variable (response, target)** | Statistics says dependent and independent; machine learning says feature and target. Show both once, then pick the course word for the session and say so. (Taught together with: independent variable.) | |
| **Dimensionality reduction** | Mixing many variables into a few that keep most of the information. | |
| **Dummy variable trap** | Keeping every level of an indicator variable makes the columns add to one. Drop one level to avoid it. | |
| **Elbow method** | Plot inertia for several k and choose the k where the drop flattens. | |
| **Ensemble** | Many small models combined into one. A forest averages trees (bagging). Boosting adds trees, each fitting the residual errors so far. | 17 |

| Term | Plain meaning | Session |
|---|---|---|
| **Epoch** | One full pass through the training records. | |
| **Error** | The gap between a prediction and the actual value for one record. | |
| **F1 score** | One number that combines precision and recall. | |
| **False positive rate** | The share of the real negatives that the model wrongly flags. It is the horizontal axis of the ROC curve. | |
| **Feature scaling** | Standardisation (z-score) puts variables on one scale. A log transform squeezes a long right tail. | 18 |
| **Feedforward neural network** | Neurons arranged in an input layer, hidden layers and an output layer, with values moving forward only. | 20 |
| **Fitted line** | The line a linear regression model draws through the points on a scatter plot. | |
| **Forecast horizon** | How far ahead the forecast looks, such as one month. | |
| **Generalisation** | How well a model works on records it has not seen. | |
| **Gradient boosting (each new tree fits the residual errors of the ensemble so far)** | Many small models combined into one. A forest averages trees (bagging). Boosting adds trees, each fitting the residual errors so far. | 17 |
| **Grid search** | Tuning that tries every combination from a list of settings (GridSearchCV). | |
| **Homoscedasticity** | The residuals have about the same spread at every predicted value. | |
| **Hyperparameter tuning (grid search, random search)** | Trying settings and choosing the best by cross-validation. | |
| **Hyperparameters (chosen)** | Settings you choose before training, such as max_depth. | |
| **Independent variable (predictor, feature)** | Statistics says dependent and independent; machine learning says feature and target. Show both once, then pick the course word for the session and say so. (Taught together with: dependent variable.) | |
| **Indicator variable (dummy variable)** | A variable that is 1 when a record belongs to a category and 0 otherwise. One-hot encoding makes one for each level. | 13 |
| **Inertia (within-cluster sum of squares)** | The sum of squared distances from each record to its cluster centre. Lower means tighter clusters. | |
| **Inference** | Using a trained model to predict for new records. | |
| **Input layer, hidden layers, output layer** | The layers of a neural network: where the data enter, the middle layers, and where the prediction leaves. | |
| **Intercept** | The coefficient is the change in the prediction for one more unit of a predictor, with the other predictors held fixed. The intercept is the prediction when every predictor is 0. | 13 |
| **K-fold cross-validation** | Split the training records into k folds, train on k minus 1, score on the held-out fold, and repeat k times. | |
| **K-means clustering** | Splits records into k clusters around k centroids. Inertia measures how tight they are. The elbow method suggests k. | 18 |
| **Lag feature** | An earlier value placed in the same row, such as last month's cases. It is built inside each district. | 19 |
| **Learning rate** | How big each correction step is in gradient descent. | |
| **Linear regression model** | A model that predicts a numeric target as an intercept plus a weighted sum of predictors. Ordinary least squares (OLS) finds the weights. | 13 |

| Term | Plain meaning | Session |
|---|---|---|
| **Log transform** | Standardisation (z-score) puts variables on one scale. A log transform squeezes a long right tail. | 18 |
| **Log-odds** | A model that bends a score from the features into a probability between 0 and 1 with the sigmoid. scikit-learn penalises large coefficients (regularisation, C = 1.0). | 15 |
| **Loss function** | A number that says how wrong the model is. Training tries to lower it. | |
| **Min-max scaling** | Rescaling a variable to run from 0 to 1. | |
| **Multilayer perceptron (MLP)** | Neurons arranged in an input layer, hidden layers and an output layer, with values moving forward only. | 20 |
| **Naive baseline** | The simplest forecast, such as repeating last year. A model must beat it. | |
| **Negative class** | Predicting one of two categories. The positive class is the outcome we look for: default = 1. The negative class is the other: repaid = 0. | 15 |
| **One-hot encoding (drop one level to avoid the dummy variable trap, perfect multicollinearity)** | Turning a text variable into one 0 or 1 indicator column per category, so a model can use it. | 15 |
| **Ordinary least squares (OLS)** | The way of fitting a line that makes the sum of squared residuals as small as possible. | |
| **Overfitting** | Overfitting: the model fits noise, so training rows score well and new rows score worse. Underfitting: too simple to see the pattern. | 17 |
| **Parameters (learned)** | The numbers a model learns from the data, such as coefficients. | |
| **Perceptron / artificial neuron** | A weighted sum of inputs plus a bias, passed through an activation function. | |
| **Permutation importance** | The drop in test score when one variable is shuffled. It shows what the model relies on. It does not show a cause. | 17 |
| **Positive class** | Predicting one of two categories. The positive class is the outcome we look for: default = 1. The negative class is the other: repaid = 0. | 15 |
| **Precision** | Precision: of the loans flagged, the share that defaulted. Recall: of the loans that defaulted, the share flagged. F1 combines both. | 16 |
| **Precision-recall curve** | Precision against recall as the decision threshold moves. Useful when the positive class is rare. | |
| **Pretrained model** | Start from a network someone else trained and train only a new last layer on your own data. | 20 |
| **Principal component analysis (PCA)** | A dimensionality reduction method: it mixes many variables into a few principal components that keep most of the variance. | 18 |
| **Principal components** | The new variables that PCA builds, ordered by how much variance they keep. | |
| **R squared (coefficient of determination)** | The share of the variation in the target that the model accounts for, from 0 to 1. | 13 |
| **Random forest (bagging, bootstrap aggregation; an ensemble)** | Many small models combined into one. A forest averages trees (bagging). Boosting adds trees, each fitting the residual errors so far. | 17 |
| **Recall (sensitivity, true positive rate)** | Precision: of the loans flagged, the share that defaulted. Recall: of the loans that defaulted, the share flagged. F1 combines both. | 16 |
| **Residual plot** | A chart of residuals against predicted values. A fair model shows an even cloud with no pattern. | |

| Term | Plain meaning | Session |
|---|---|---|
| **Residuals (actual minus predicted)** | Actual value minus predicted value for each record. | |
| **RFM analysis** | Three numbers per customer: recency (days since the last order), frequency (number of orders) and monetary value (total spend). | 18 |
| **ROC curve** | The ROC curve plots the true positive rate (recall, sensitivity) against the false positive rate at every threshold. AUC, the area under it, is 0.5 for a coin flip and 1.0 for a perfect model. | 16 |
| **Root node, internal node, leaf node, split** | The parts of a decision tree: the first question, later questions, the final answers, and the rule that sends records left or right. | |
| **Scatter plot** | A chart with one dot per record, one variable on each axis. | |
| **Seasonal naive forecast (a naive baseline)** | The simplest forecast: copy the same month from last year. A model must beat it to be worth using. | 19 |
| **Sigmoid (logistic) function** | The S-shaped curve that turns any score into a probability between 0 and 1. | |
| **Standardisation (z-score)** | Rescaling a variable to mean 0 and standard deviation 1 (z-score). | |
| **Stratified k-fold** | K-fold in which every fold keeps the same share of each class. | |
| **Test set** | The training set is what the model learns from. The test set is held back and used once at the end to score the model. | 14 |
| **Time-based split (temporal split)** | Train on the months before a cut date and test on the months after it. Never split a time series at random. | 19 |
| **Training by gradient descent** | Repeated small corrections to the weights that lower the loss function, using backpropagation to find each correction. | |
| **Training set** | The training set is what the model learns from. The test set is held back and used once at the end to score the model. | 14 |
| **Transfer learning** | Start from a network someone else trained and train only a new last layer on your own data. | 20 |
| **True positive, false positive, true negative, false negative** | The four cells of a confusion matrix: right yes, wrong yes, right no, wrong no. | |
| **True positive rate (recall, sensitivity)** | The share of the real positives that the model flags. It is the vertical axis of the ROC curve. | |
| **Underfitting** | Overfitting: the model fits noise, so training rows score well and new rows score worse. Underfitting: too simple to see the pattern. | 17 |
| **Validation set (hold-out)** | Records held out to compare models and settings before the final test. | |
| **Walk-forward validation** | Train on the past, test on the next period, then move the window forward and repeat. | |

## Text, language models, delivery and governance

| Term | Plain meaning | Session |
|---|---|---|
| **Audit log (prompt log)** | A record of what the assistant was asked and what it answered, kept so people can check. | |
| **Bag of words (count vectorizer)** | Bag of words counts each word in each message and ignores the order. TF-IDF weights those counts so rare words count more. | 21 |
| **Branch** | A separate line of work in a repository. | |
| **Chunk (chunking)** | A piece of a document, small enough to search and to fit in a prompt. | |

| Term | Plain meaning | Session |
|---|---|---|
| **Chunking** | Cutting documents into chunks before they are searched. | |
| **Commit** | A saved snapshot of the project with a short message. | |
| **Concept drift** | The relationship between the inputs and the outcome changes over time. | |
| **Context window** | A token is a word or piece of a word that a language model reads and writes. The context window is how many tokens it can read at once. | 22 |
| **Cosine similarity** | A score from -1 to 1 for how alike two vectors are. Semantic search ranks messages by that score. | 21 |
| **Data controller** | Information that identifies a person. The person is the data subject. The organisation that decides how it is used is the data controller. | 23 |
| **Data drift (and concept drift when the relationship changes)** | Data drift: the inputs change after training. Concept drift: the link between the inputs and the outcome changes. | 23 |
| **Data subject (Uganda Data Protection and Privacy Act, 2019)** | Information that identifies a person. The person is the data subject. The organisation that decides how it is used is the data controller. | 23 |
| **Deploy** | Put the app on the internet so other people can use it. | |
| **Deployment** | The act of putting a model or app where users can reach it. | |
| **Disparate impact** | Compare approval and error rates across groups. A proxy variable stands in for a column you left out. Disparate impact is the ratio of two approval rates. | 23 |
| **Embedding (a vector)** | A list of numbers that stands for the meaning of a message. Messages with similar meanings get similar vectors. | 21 |
| **Evaluation set (a labelled test set)** | A labelled list of test questions with the right passage. Precision at k and recall at k score the search. | 22 |
| **Executive summary** | A short opening paragraph with the answer, the evidence, the main limit and the decision you ask for. A busy reader may read only this. | 24 |
| **Fairness audit (compare error and approval rates across groups; disparate impact)** | Compare approval and error rates across groups. A proxy variable stands in for a column you left out. Disparate impact is the ratio of two approval rates. | 23 |
| **Few-shot** | The instruction you give a model. A system prompt sets rules. Zero-shot gives no examples. Few-shot gives a few worked examples. | 22 |
| **Git repository** | A project folder with its full history of changes. | |
| **GitHub** | A website that hosts Git repositories. | |
| **Hallucination** | Fluent text that no source supports. It can read well and still be false. | 22 |
| **Model card** | A one-page note that says what a model is for, what data it used, how well it works and where it fails. | |
| **Model serialisation (persistence)** | Saving a trained model to a file, for example with joblib or pickle, so it can be loaded later without training again. | 23 |
| **Personal data** | Information that identifies a person. The person is the data subject. The organisation that decides how it is used is the data controller. | 23 |
| **Precision at k** | A labelled list of test questions with the right passage. Precision at k and recall at k score the search. | 22 |
| **Prompt (a system prompt sets rules; zero-shot and few-shot prompting)** | The instruction you give a model. A system prompt sets rules. Zero-shot gives no examples. Few-shot gives a few worked examples. | 22 |

| Term | Plain meaning | Session |
|---|---|---|
| **Proxy variable** | Compare approval and error rates across groups. A proxy variable stands in for a column you left out. Disparate impact is the ratio of two approval rates. | 23 |
| **Push** | Send your commits to the copy on GitHub. | |
| **Recall at k** | A labelled list of test questions with the right passage. Precision at k and recall at k score the search. | 22 |
| **Regular expression (regex)** | A pattern that finds text, such as every phone number in a message. | |
| **Reproducibility (fix the random seed, record library versions, requirements file)** | A colleague who runs your notebook gets your result. It needs a fixed random seed, recorded library versions and a run from the top. | 23 |
| **Requirements file** | A text file (requirements.txt) listing the libraries and versions the project needs. | |
| **Retrieval-augmented generation (RAG)** | Find the relevant chunks of your documents first (retrieval and ranking), then ask the model to answer from them (grounding). | 22 |
| **Rubric** | A list of the qualities that good work has, one row for each. You use it on your own project and to give a partner feedback. Nothing is scored. | 24 |
| **Seed** | The starting number of a random generator. Fixing it gives the same result every run. | |
| **Semantic search** | A score from -1 to 1 for how alike two vectors are. Semantic search ranks messages by that score. | 21 |
| **Sentiment analysis (text classification)** | A supervised model that gives each message a label, such as positive, neutral or negative. | 21 |
| **Stop words** | Cleaning raw text into tokens: lowercase it, remove noise with a regular expression, split it into words and drop filler words. | 21 |
| **System prompt** | The instruction you give a model. A system prompt sets rules. Zero-shot gives no examples. Few-shot gives a few worked examples. | 22 |
| **Text preprocessing** | Cleaning raw text into tokens: lowercase it, remove noise with a regular expression, split it into words and drop filler words. | 21 |
| **TF-IDF (term frequency, inverse document frequency)** | Bag of words counts each word in each message and ignores the order. TF-IDF weights those counts so rare words count more. | 21 |
| **Token** | A token is a word or piece of a word that a language model reads and writes. The context window is how many tokens it can read at once. | 22 |
| **Tokenisation** | Cleaning raw text into tokens: lowercase it, remove noise with a regular expression, split it into words and drop filler words. | 21 |
| **Vector** | A list of numbers that stands for the meaning of a message. Messages with similar meanings get similar vectors. | 21 |
| **Versions** | The numbers of the libraries you used. Record them so others can rebuild your work. | |
| **Zero-shot** | The instruction you give a model. A system prompt sets rules. Zero-shot gives no examples. Few-shot gives a few worked examples. | 22 |